

MITRE ATLAS Framework

CompTIA SecAI+: AI Security Certification Complete Exam Prep 2026 | Section 10: AI Threat Modeling Resources

1. What Is MITRE ATLAS?

MITRE ATLAS—Adversarial Threat Landscape for Artificial-Intelligence Systems—is a knowledge base that documents how adversaries attack AI systems using real-world tactics, techniques, and procedures. Developed by MITRE, the organization behind the widely used ATT&CK framework for traditional cybersecurity, ATLAS applies the same adversary-centric approach to AI.

The critical distinction between ATLAS and other AI risk frameworks is perspective. OWASP Top 10 lists and the MIT AI Risk Repository describe what can go wrong from a risk management standpoint. ATLAS describes how attackers actually make it go wrong—the objectives they pursue, the techniques they use, and the sequence in which they typically operate. This adversary-centric viewpoint is essential for red team exercises, penetration testing, and threat modeling that needs to reflect real-world attack patterns rather than theoretical risk categories.

If you already work with MITRE ATT&CK for traditional infrastructure, ATLAS will feel immediately familiar. The same tactic-and-technique taxonomy applies, but the subject matter is AI systems: machine learning models, training pipelines, inference services, and the data that powers them.

2. ATLAS Tactics: The Attack Lifecycle

ATLAS organizes AI attacks into tactics—high-level objectives that represent what the attacker is trying to accomplish at each stage of an attack campaign. Understanding tactics allows security teams to reason about complete attack scenarios rather than isolated techniques.

Tactic	Attacker Objective	Defender Priority
Reconnaissance	Gather intelligence about the AI system and environment	Limit public exposure of AI technology and architecture details
Resource Development	Build tools, infrastructure, or capabilities needed for the attack	Monitor for adversary infrastructure targeting your AI stack
Initial Access	Establish a foothold in the target environment	Strong authentication, least-privilege access, API security
ML Model Access	Gain direct or indirect interaction with the target model	Rate limiting, authorization controls, anomaly detection on queries
Execution	Run adversarial code or queries to interact with the AI system	Input validation, sandboxing, query monitoring
Persistence	Maintain long-term access or influence over the AI system	Integrity monitoring, model versioning, access reviews
Defense Evasion	Avoid detection while conducting malicious activities	Adversarial training, layered detection, output monitoring
Exfiltration	Steal models, training data, or sensitive information	DLP, encryption, access logging, API rate limiting

Impact	Degrade, disrupt, or corrupt the AI system or its outputs	Redundancy, integrity verification, incident response planning
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Each tactic contains multiple techniques—specific methods an attacker can use to achieve that tactic's objective. The combination of tactics and techniques allows security professionals to model complete attack campaigns and design controls that interrupt them at multiple stages.

3. Reconnaissance: How Attackers Study AI Systems

Almost every sophisticated attack begins with reconnaissance. Before launching any attack, adversaries invest time understanding the target AI system—its technology, architecture, data sources, and deployment environment. Every piece of information gathered narrows the attack surface and increases the probability of a successful outcome.

ATLAS documents multiple reconnaissance techniques relevant to AI systems. Attackers may search public repositories such as GitHub, Hugging Face, or academic publication archives to identify which frameworks an organization uses. Job postings often reveal the ML stack and team structure. Exposed APIs, published research papers, conference presentations by the organization's engineers, and inadvertently public model endpoints can all yield intelligence about the AI system's internals.

Defensive countermeasures at this stage focus on reducing unnecessary information exposure: reviewing public documentation before release, enforcing strict access controls on APIs and model endpoints, monitoring for unusual discovery activities, and training engineers to be thoughtful about what technical details are shared publicly. A well-informed attacker is a more dangerous attacker.

4. ML Model Access: Controlling the Gateway

Key Principle: Many of the most damaging AI attacks—inference attacks, adversarial example crafting, model extraction—require the attacker to interact directly with the model. Controlling who can access the model and under what conditions is therefore one of the highest-value defensive investments available to AI security teams.

MITRE ATLAS identifies multiple pathways attackers use to gain model access. These include compromising legitimate user accounts through phishing or credential theft, exploiting authentication or authorization flaws in inference APIs, leveraging insider access, exploiting software vulnerabilities in the serving infrastructure, or simply finding publicly exposed endpoints that were never intended to be accessible.

Defensive measures at this stage include multi-factor authentication, role-based access control with minimal permissions, API key management and rotation, rate limiting on inference requests, monitoring for unusual query patterns, and regular access reviews to revoke unnecessary permissions. Each of these controls either prevents unauthorized access or creates observable signals when access is attempted inappropriately.

5. Inference Attacks: Extracting Secrets Through Queries

Once an attacker has access to a deployed model, they can begin extracting information through careful interaction without ever compromising the underlying infrastructure. These inference attacks exploit the model's outputs as an information channel.

- **Model Inversion:** by submitting carefully chosen inputs and analyzing outputs, attackers attempt to reconstruct sensitive training data that the model has memorized, such as individual records in a medical training dataset.
- **Membership Inference:** the attacker determines whether a specific data record was included in the training dataset, which can reveal whether a person's data was used without their consent.
- **Model Extraction:** through repeated queries, the attacker builds a functionally equivalent replica of the target model, stealing the intellectual property embedded in the model's learned parameters without ever accessing the model file directly.

Defending against inference attacks requires limiting the information available through model responses (reducing output confidence scores, for example), applying rate limiting to prevent the volume of queries needed for extraction, monitoring for statistical patterns in query sequences that suggest systematic probing, and applying privacy-preserving techniques such as differential privacy during training.

6. Real-World Incident: Adversarial Evasion in Practice

Real-World Incident — Adversarial Patch Attack on Computer Vision (2019): Researchers at Carnegie Mellon University demonstrated that a small printed patch—when attached to a person's clothing—could cause state-of-the-art AI object detection systems to fail to identify that person as a human being. The attack worked consistently across different viewing angles and lighting conditions. This incident documented in MITRE ATLAS under the Defense Evasion tactic illustrates how adversarial inputs that appear meaningless to humans can systematically fool deployed AI systems. Security AI that relies on computer vision, such as physical access control or video surveillance analytics, faces identical vulnerabilities.

This incident demonstrates why ATLAS's adversary-centric perspective is valuable. The risk frameworks might describe adversarial examples as a known risk category. ATLAS shows specifically how that technique is constructed and applied in practice—knowledge that is essential for designing effective countermeasures.

7. Poisoning: Corrupting the Foundation

Data poisoning attacks target AI systems before they are deployed. Rather than attacking a running model, the adversary corrupts the training data so that the model learns flawed patterns—patterns that serve the attacker's objectives. Because the corruption becomes embedded in the model's weights during training, its effects persist throughout the model's operational lifetime.

ATLAS documents poisoning as a pre-deployment tactic with significant post-deployment consequences. An attacker who can inject malicious samples into a training dataset—by compromising a data collection system, exploiting a public dataset that the organization downloads without verification, or performing a

supply chain attack on a third-party data provider—can influence the model's behavior in precise and targeted ways.

Defending against poisoning requires treating training data with the same security rigor applied to code: verifying the provenance of every data source, applying integrity checks to datasets, monitoring for anomalies in training data distributions, maintaining audit logs of all data ingestion events, and validating model behavior against expected performance benchmarks before promoting any model to production.

8. Model Backdoors: Hidden Triggers

Model backdoors represent a particularly insidious form of poisoning. Rather than degrading overall model performance, a backdoor embeds hidden behavior that activates only when a specific trigger input pattern is encountered. Under normal operating conditions, the backdoored model performs correctly and passes standard evaluation tests. The attack remains invisible until the trigger is presented.

Consider a security application: an AI-powered intrusion detection system that correctly identifies virtually all network attack patterns—except one specific signature that the attacker embedded as the backdoor trigger. The system passes all quality assurance tests because normal attacks are detected normally. Only the attacker's specific traffic pattern is misclassified as benign.

Detecting backdoors is significantly harder than detecting general performance degradation. Defenses include using trusted, audited training pipelines with no external modification points, conducting diverse adversarial testing scenarios designed to probe for unexpected behavior, applying neural cleanse or similar backdoor detection techniques to trained models, and maintaining strict version control over model artifacts so that any unexpected changes are detectable.

9. Exfiltration and Impact: Completing the Attack

After reconnaissance, access establishment, and model interaction, many attackers turn to stealing AI assets or disrupting AI system function. ATLAS maps both of these objectives explicitly.

Attack Objective	ATLAS Tactic	What Attackers Target	Defensive Priority
Steal AI assets	Exfiltration	Trained model files, proprietary training datasets, threat intelligence data	Encryption, access controls, data loss prevention, API rate limiting
Disrupt AI function	Impact	Model integrity, inference service availability, prediction reliability	Redundancy, integrity verification, incident response, rate limiting

For security AI specifically, impact attacks carry compounded risk: a compromised or unavailable AI detection system weakens the organization's defensive posture precisely when an attacker may be preparing a second, more damaging attack. Treating AI security systems as critical infrastructure—with the same availability and integrity protections applied to other critical assets—is essential.

10. Analogy: ATLAS as a Chess Clock

Memory Anchor — ATLAS as a Chess Clock: Chess masters study their opponents' known opening lines not because the opponent will follow a script, but because pattern recognition built on prior games enables faster and better decisions under pressure. MITRE ATLAS works the same way. Each documented tactic and technique represents a move that real attackers have made against real AI systems. Studying those moves does not tell you exactly what the next attack will look like, but it dramatically shortens the time between observing attacker behavior and recognizing what stage of the attack lifecycle is underway.

11. Using ATLAS Defensively and in Red Teaming

Although ATLAS documents attacker behavior, its primary intended use is defensive. Security teams apply ATLAS in two main ways: structured threat modeling and red team exercises.

- **Threat modeling with ATLAS:** review each ATLAS tactic and ask whether the documented techniques could be applied to your AI system. For each applicable technique, assess existing controls and identify gaps. This produces a prioritized defensive roadmap grounded in real attacker behavior rather than theoretical risk categories.
- **Red team exercises with ATLAS:** security professionals simulate documented ATLAS techniques against the organization's AI systems to evaluate whether existing controls detect and respond to realistic attacks. Red teaming with ATLAS ensures exercises reflect actual adversary behavior rather than generic penetration testing scenarios.

Looking at security from the attacker's perspective consistently reveals defensive gaps that traditional risk-based reviews overlook. This is the core contribution of ATLAS: not just knowing that a risk exists, but understanding the specific methods attackers use to exploit it.

12. ATLAS, OWASP, and MIT Repository: Complementary Roles

MITRE ATLAS does not exist in isolation. It is most powerful when used alongside the OWASP ML Top 10, OWASP LLM Top 10, and the MIT AI Risk Repository—each of which contributes a distinct perspective to AI security.

Framework	Primary Lens	What It Adds to the Picture
OWASP ML Top 10	Risk prioritization	The most common and impactful ML security risks — where to start
OWASP LLM Top 10	LLM-specific risk prioritization	Critical vulnerabilities unique to language model deployments
MIT AI Risk Repository	Comprehensive risk breadth	Hundreds of AI risks across technologies, domains, and lifecycle stages
MITRE ATLAS	Adversary behavior	How attackers actually execute attacks — tactics, techniques, real cases

In practice: use OWASP to prioritize your highest-risk starting points, use the MIT repository to ensure you have not missed less common risks, and use ATLAS to understand specifically how those risks are exploited in practice. Together, these frameworks provide the coverage, priority, and adversary perspective needed for comprehensive AI security planning.

Key Terms Glossary

Term	Definition
MITRE ATLAS	Adversarial Threat Landscape for Artificial-Intelligence Systems; a knowledge base of adversary tactics, techniques, and procedures targeting AI systems.
Tactic	A high-level attacker objective in the ATLAS framework, such as Reconnaissance, ML Model Access, or Defense Evasion.
Technique	A specific method an attacker uses to achieve a tactic's objective, documented with real-world examples in ATLAS.
Reconnaissance	The ATLAS tactic covering intelligence gathering about the target AI system before the attack begins.
ML Model Access	The ATLAS tactic covering how adversaries gain interaction with the target machine learning model.
Inference Attack	A class of attacks that extract information from a model through carefully crafted queries, including model inversion, membership inference, and model extraction.
Model Extraction	An inference attack in which an adversary reconstructs a functional copy of the target model through repeated queries.
Data Poisoning	An attack that manipulates training data to alter the model's learned behavior, embedding vulnerabilities before deployment.
Model Backdoor	A poisoning attack that embeds hidden behavior in a model that activates only when a specific trigger input is present.
Adversarial Example	A carefully crafted input that appears normal but causes an AI model to produce incorrect outputs, used primarily for evasion attacks.
MITRE ATT&CK	The foundational MITRE knowledge base documenting adversary tactics and techniques for traditional information systems; ATLAS is its AI-focused counterpart.

Quick Revision Checklist

- MITRE ATLAS is the AI-focused counterpart to MITRE ATT&CK, documenting how adversaries attack AI systems.
- ATLAS organizes attacks into tactics (objectives) and techniques (specific methods), covering the full AI attack lifecycle.
- Reconnaissance reduces attacker uncertainty — limit public exposure of AI architecture and technology details.
- ML Model Access is the gateway tactic — most sophisticated AI attacks require direct model interaction to execute.
- Inference attacks (model inversion, membership inference, model extraction) extract information through the model's outputs.
- Adversarial examples crafted for defense evasion can cause deployed AI models to misclassify inputs consistently.

- Data poisoning and model backdoors corrupt AI behavior before deployment and persist throughout the model's lifetime.
- Exfiltration targets trained models and datasets as valuable intellectual property; Impact attacks degrade AI availability and integrity.
- Use ATLAS for threat modeling (identify applicable techniques, map controls, identify gaps) and red team exercises.
- ATLAS complements OWASP frameworks (priority) and the MIT repository (breadth) — use all three together for complete coverage.

10 Exam-Based MCQs — MITRE ATLAS Framework

Test yourself after reading. These questions are written in real exam style — scenario-heavy, with plausible distractors. Read every explanation, including for questions you answered correctly.

Q1. Scenario: *The security team at a financial institution has deployed an AI-powered fraud detection model. The CISO has asked the red team to simulate a realistic advanced persistent threat campaign targeting the AI system using the MITRE ATLAS framework as their attack playbook. Which MITRE ATLAS tactic should the red team focus on FIRST to assess whether the organization's defensive controls are effective at preventing sophisticated AI attacks?*

- A) Impact, because degrading AI function is the ultimate attacker objective
- B) Reconnaissance, because most sophisticated attacks begin with intelligence gathering
- C) Exfiltration, because model theft is the most common AI attack outcome
- D) Defense Evasion, because adversarial examples are the best-known AI attack technique

Answer: B **Explanation:** B is correct because ATLAS attack campaigns typically begin with Reconnaissance, and disrupting the attacker's ability to gather intelligence is one of the earliest and most cost-effective defensive interventions. A is wrong because Impact is the final attack stage, not the starting point for defensive assessment. C is wrong because Exfiltration is a later-stage tactic and not necessarily the most common starting focus. D is wrong because Defense Evasion cannot occur until the attacker has already completed Reconnaissance and gained Model Access.

Q2. Scenario: *A security architect is designing controls for a healthcare organization's AI diagnostic assistant that classifies patient imaging results. The CISO has reviewed the MITRE ATLAS framework and wants to ensure the team addresses the ML Model Access tactic specifically. Which defensive control BEST addresses the MITRE ATLAS ML Model Access tactic?*

- A) Adversarial training of the deployed model
- B) Strong authentication, authorization, and rate limiting on inference APIs
- C) Encrypting training data at rest
- D) Publishing model architecture details to improve community oversight

Answer: B **Explanation:** B is correct because ML Model Access is achieved through authentication or authorization weaknesses, exposed APIs, or stolen credentials; strong authentication, least-privilege access, and rate limiting directly counter these pathways. A is wrong because adversarial training addresses Defense Evasion, not initial model access. C is wrong because encrypting training data at rest protects

against data theft but does not prevent unauthorized model query access. D is wrong because publishing architecture details increases reconnaissance effectiveness, which is the opposite of a defensive measure.

Q3. Scenario: A threat intelligence analyst at a cybersecurity company notices that a competitor's public machine learning API has received an unusually high volume of queries from a cluster of IP addresses over the past three weeks. The query patterns appear to be systematically varied rather than random. An attacker submits tens of thousands of carefully varied queries to a competitor's deployed machine learning API over several weeks. What ATLAS technique is MOST likely being executed?

- A) Data Poisoning — the attacker is corrupting the model's training data
- B) Model Backdoor — the attacker is embedding a hidden trigger in the model
- C) Model Extraction — the attacker is building a functional replica of the model through systematic querying
- D) Reconnaissance — the attacker is gathering information about the organization

Answer: C **Explanation:** C is correct because Model Extraction involves submitting large volumes of queries to reconstruct a functional copy of the target model, which matches the described behavior of systematic, varied API queries over an extended period. A is wrong because Data Poisoning requires access to the training pipeline, not the inference API. B is wrong because backdoors are embedded during training and require training data or pipeline access, not inference-time queries. D is wrong because reconnaissance involves information gathering about the system's characteristics, not the systematic query volume described.

Q4. Which of the following BEST describes a Model Backdoor attack in the context of MITRE ATLAS?

- A) An attack that causes denial of service by overwhelming the inference service with queries
- B) A poisoning attack that embeds hidden behavior activated only by a specific trigger input
- C) An attack that extracts the model's training data through repeated output analysis
- D) A vulnerability in the model serving infrastructure that allows remote code execution

Answer: B **Explanation:** B is correct because a model backdoor is a specific form of poisoning attack that embeds covert functionality: the model behaves normally in all circumstances except when a specific trigger pattern is present, at which point it produces attacker-controlled outputs. A is wrong because overwhelming inference with queries describes a denial-of-service impact attack, not a backdoor. C is wrong because extracting training data describes a Model Inversion inference attack. D is wrong because infrastructure vulnerabilities are software security issues, not AI-specific backdoor attacks.

Q5. Which primary distinction separates MITRE ATLAS from the OWASP ML Top 10 as threat modeling resources?

- A) ATLAS covers only LLM vulnerabilities while OWASP covers all machine learning risks
- B) ATLAS documents adversary tactics and techniques while OWASP prioritizes the most critical risks
- C) ATLAS provides compliance frameworks while OWASP focuses on attack simulation

D) ATLAS is maintained by academic researchers while OWASP is maintained by government agencies

Answer: B **Explanation:** B is correct because ATLAS takes an adversary-centric perspective, documenting the tactics and techniques attackers use, while OWASP provides a prioritized risk list identifying the most critical vulnerabilities. These are complementary, not competing, perspectives. A is wrong because ATLAS covers AI systems broadly, including traditional ML and LLMs; OWASP has separate lists for ML and LLMs. C is wrong because neither framework is primarily a compliance framework; both are security knowledge bases. D is wrong because MITRE is not an academic institution, and OWASP is not a government agency.

Q6. An organization discovers that its AI malware detection system consistently misclassifies one specific malware family as benign, even though it correctly identifies all other malware. What attack is MOST likely responsible?

- A) Model Extraction — a competitor has stolen the model's weights
- B) Membership Inference — an attacker determined which samples were in the training set
- C) Model Backdoor — an attacker embedded a trigger during training that causes misclassification of a specific pattern
- D) Denial of Service — the inference service is being overwhelmed with queries

Answer: C **Explanation:** C is correct because the described behavior—consistent correct classification of all threats except one specific pattern—is the hallmark of a model backdoor attack, where the backdoor trigger causes misclassification only for that specific input pattern. A is wrong because model extraction creates a copy of the model and does not alter the original model's behavior. B is wrong because membership inference determines whether data was in the training set and does not alter model predictions. D is wrong because denial of service affects availability, not selective misclassification of specific inputs.

Q7. Which ATLAS tactic is MOST directly addressed by validating training dataset provenance and applying integrity checks to data before training?

- A) Reconnaissance
- B) Defense Evasion
- C) Exfiltration
- D) Poisoning (addressed through the Initial Access and Execution tactics targeting training data)

Answer: D **Explanation:** D is correct because data poisoning attacks manipulate training data; validating data provenance and applying integrity checks before training directly defends against adversaries injecting malicious samples into the training pipeline. A is wrong because Reconnaissance involves information gathering before any attack on the system, not training data manipulation. B is wrong because Defense Evasion covers techniques used during deployment, not during the training data ingestion phase. C is wrong because Exfiltration involves stealing model or data assets after they exist, not corrupting them during training.

Q8. Why does the MITRE ATLAS framework document real-world case studies alongside tactics and techniques?

- A) To provide legal documentation for prosecution of AI attackers
- B) To demonstrate that AI attacks are theoretical and rarely occur in practice
- C) To show how individual techniques combine into complete attack campaigns observed against real systems
- D) To provide an exhaustive list of all known AI vulnerabilities

Answer: C **Explanation:** C is correct because real-world case studies in ATLAS show how tactics and techniques are sequenced and combined in actual attack campaigns, transforming abstract technique descriptions into practical lessons about adversary behavior. A is wrong because ATLAS is a security knowledge base, not a legal documentation system. B is wrong because the case studies demonstrate the opposite: AI attacks occur in practice and have affected real organizations. D is wrong because ATLAS documents adversary behavior, not a comprehensive vulnerability catalog; that is closer to the role of the MIT AI Risk Repository.

Q9. What makes model backdoor attacks particularly difficult to detect compared to other AI attacks?

- A) They require physical access to the data center hosting the model
- B) The model performs correctly under normal conditions and only exhibits malicious behavior when the trigger is present
- C) They always cause measurable performance degradation across all inputs
- D) They can only be executed by insiders with direct model file access

Answer: B **Explanation:** B is correct because a backdoored model passes normal quality assurance testing by performing correctly on all standard inputs; only the specific trigger pattern activates the malicious behavior, making the attack invisible until the trigger is deliberately or accidentally encountered. A is wrong because backdoor attacks are embedded during training and do not require physical access to detect or exploit. C is wrong because backdoors specifically avoid causing general performance degradation, which is what makes them difficult to detect. D is wrong because backdoors can be introduced through supply chain attacks on training data without requiring direct model file access by the attacker.

Q10. A security team wants to use MITRE ATLAS to test whether their AI system can withstand real-world attacks. What is the MOST appropriate methodology?

- A) Review ATLAS tactics theoretically and document which ones could theoretically apply
- B) Use ATLAS techniques as the basis for structured red team exercises that simulate documented adversary behavior
- C) Implement all ATLAS-documented controls simultaneously and verify that no attacks succeed
- D) Share ATLAS documentation with the AI development team and ask them to self-assess security

Answer: B **Explanation:** B is correct because ATLAS red team exercises simulate documented adversary techniques against the actual AI system, producing empirical evidence of what controls work and where gaps exist. This is significantly more reliable than theoretical assessment or self-assessment. A is wrong

because purely theoretical review does not reveal whether actual controls withstand attack; empirical testing is required. C is wrong because ATLAS documents attacker techniques, not defensive controls; and simultaneously implementing all possible controls is neither practical nor methodologically sound. D is wrong because self-assessment by development teams lacks the independence and adversarial perspective that red teaming provides.